

to prevent injuries when players are about to exceed their physical thresholds.

Other technologies to support intelligent decision-making in sports

Helmet cams and lasers are used to compliment run-out decisions in the game of cricket. Sensors in balls and in the boundary rope are used to determine if the batsman is out or has hit a boundary (four runs). A stump camera is used to determine close catching decisions. Umpires need feedback about whether a front-foot no ball has been made. It can be in the form of a beeping sound if the bowler crosses the popping crease. This helps them focus on what the batsman and fielders are doing.

Inertial sensors have been developed to test a bowling action during matches to see if the bowlers are bending their arm too much during delivery. These sensors are lightweight, cost-effective and wearable on the bowler's arm. These do not hinder the bowling performance while still allow the information about the bowling action to be assessed in near real time in both match and training environments.

Benefits of using technology in sports

Rules enforcement and referee assistance. Decision referral systems, Hawk Eye, hot-spot, shot-tracer, radar guns and snick-o-meter help administrators deliver a fair and an unbiased experience in sports.

Training and coaching support. Wearables and electronic devices can be attached to an athlete's body, clothing, footwear or equipment to track critical information such as heartrate, speed, distance and so

on. This, in turn, can be used for the athlete's development.

Media and reach. The Internet, high-speed cables, apps, HDTVs, blogs, social media platforms and the like have brought sports deep into our everyday lives, and have created a wider connect between sports and their fans.



Snick-o-meter for no bat to ball hit (Credit: <https://sportsmatik.com>)



Snick-o-meter for ball to bat hit (Credit: <https://scroll.in>)

Improved audience experience. Organisers are working to deliver an enriched audience experience through live HD screens, services on order and other in-stadium engagement services.

Better sports equipment and clothing. Improvements in the design of sports equipment and apparels have improved almost every single aspect for athletes. Sporting equipment is continually undergoing research and development to improve performance. Stronger and lighter steel is being used in the manufacturing of sails, bicycle tyres, football boots, tennis rackets, helmets, body armour and more.

Raving cycles and rowing shells made of lightweight but strong materials minimise the drag through air or water.

The development of sports equipment and apparels is an industry in itself, as it creates an opportunity for investment and employment. In the context of professional sports,

not only vital data of athletes but also environmental data is relevant. It is therefore not surprising that all other physical objects (such as bats, gloves, balls or even the floor) are increasingly being enriched with sensors that collect data and provide insights to players, coaches, fans and regulators.

Future of technology in sports

As technology adoption improves in India, the next stage of sport-tech will be the use of different platforms to deliver integrated data and analytics to enhance the total sports experience. New technologies will spot doping and cheating, improve athlete safety and timely injury prevention, and prevent over exhaustion.

Real-time data and information will become available for crowd management, coaching programmes and training schedules. Statistical data will drive decisions on scoring patterns, team and athlete performance.

With changing shape and materials used in sports, like tennis rackets and football boots, technology has changed every dimension of the sports industry. Innovations have made sports safer and more accurate. But caution must be paid to ensure that sports remain accessible and fair. The challenge lies in ensuring that technology facilitates sports and not overshadow the core skills of the athletes. **EFY**